CSA47 - Deep Learning

Assignment 1: CO1 - Mathematical Foundations and Machine Learning Principles

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Assignment	CO1 - Mathematical Foundations and ML Principles (Healthcare Risk Prediction), 50 marks

ASSIGNMENT 1 - CO1: Mathematical Foundations and Machine Learning Principles

Notation. X is the feature matrix (rows = patients, columns = features), x_i is the feature vector of patient i, y is the label, P(A) is probability, H(Y) is entropy, and d(x,z) is distance.

Task 1 - Patient Data Representation and Normalization (5 marks)

1(a) Effect of raw, uncentred features. Age (about 0-100 years), blood pressure (about 90-180 mmHg), glucose (about 70-300 mg/dL) and cholesterol (about 120-300 mg/dL) have very different spreads. Without scaling:

- **Distance dominance.** In Euclidean distance (used by K-NN and K-means), glucose and cholesterol contribute most of the squared difference, so age is effectively ignored when judging patient similarity.
- **Poor optimization.** The loss surface becomes elongated: gradients are large along high-range features and small along low-range ones. Gradient descent zig-zags and needs a smaller learning rate.
- **Unfair regularization.** An L2 penalty shrinks weights on large-unit features and small-unit features by different amounts purely because of units, not clinical importance.
- Tree-based models are much less affected, because splits on a threshold are unchanged by monotonic rescaling.

1(b) Feature matrix and min-max normalization. Three hypothetical patients:

Patient	Age (years)	BP (mmHg)	Glucose (mg/dL)	Cholesterol (mg/dL)
P1	35	120	90	180
P2	60	150	140	240
P3	45	130	110	200

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X = [[35, 120, 90, 180], [60, 150, 140, 240], [45, 130, 110, 200]] shape 3 x 4
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Min-max scaling is applied column by column: x' = (x - min_j) / (max_j - min_j).

Feature	min	max	range
Age	35	60	25
BP	120	150	30
Glucose	90	140	50
Cholesterol	180	240	60

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P1: (0, 0, 0, 0) P2: (25/25, 30/30, 50/50, 60/60) = (1, 1, 1, 1) P3: (10/25, 10/30,
20/50, 20/60) = (0.40, 0.33, 0.40, 0.33)
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Verification: every normalized value lies in [0, 1], and each column has minimum 0 and maximum 1. A standardized alternative z = (x - mu)/sigma gives mean 0 and standard deviation 1, but mu and sigma must be computed on the training split only and reused unchanged for validation, test and future data.

1(c) Why scaling matters.

- **Distance-based learning.** Suppose two patients differ by 10 years of age and 20 mg/dL of glucose. The raw squared contributions are $10^2 = 100$ and $20^2 = 400$, so glucose dominates purely through units. After min-max scaling with ranges of 25 years and 50 mg/dL, the same differences become 0.40 and 0.40, so both count equally in the metric.
- **Gradient-based learning.** A gradient step is theta(t+1) = theta(t) - eta * grad L(theta(t)). Scaled inputs give a more uniform curvature, so a single learning rate works across parameters and convergence is faster and more stable.
- **Good practice.** Fit scaling statistics on training data only; handle impossible values and missing data first; watch for outliers, since min-max scaling compresses most data when extremes are present, and use robust scaling for skewed clinical measurements. Equal numerical range does not mean equal clinical importance.

Task 2 - Vector Representation and Matrix Operations (2 marks)

2(a) Vector notation. Use the order age, glucose, blood pressure, cholesterol. The patient is $x = [x_{age}, x_{glu}, x_{BP}, x_{chol}]^T$ and the importance weights are $w = [w_{age}, w_{glu}, w_{BP}, w_{chol}]^T$. The weighted score is $s = w^T x$, or $s = w^T x + b$ with an intercept. For n patients stacked as rows of X, all scores are computed at once as $s = Xw$.

2(b) Numerical example. Normalized patient $x = [0.45, 0.70, 0.55, 0.60]^T$, weights $w = [0.20, 0.40, 0.20, 0.20]^T$ (weights sum to 1):

```
s = 0.20(0.45) + 0.40(0.70) + 0.20(0.55) + 0.20(0.60) = 0.09 + 0.28 + 0.11 + 0.12 = 0.60
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The score is 0.60 on a constructed 0-1 scale. Glucose contributes most (0.28, about 47% of the score) because it has the largest weight, not because it is intrinsically most important. The score is not a probability until it is calibrated, and the weights should be learned from representative data or agreed through a validated clinical protocol.

Task 3 - Probability and Conditional Probability (6 marks)

Let D = has the disease and G = has elevated glucose. Counts from the sample of 1,000 patients:

Group	Elevated glucose (G)	Not elevated	Total
Disease (D)	160	40	200
No disease (not D)	120	680	800
Total	280	720	1000

(a) P(G) = 280 / 1000 = 0.28 (28%). This is the marginal probability and ignores disease status.

(b) P(D | G) = P(D and G) / P(G) = (160/1000) / (280/1000) = 160/280 = 0.571 (57.1%).

(c) P(G | D) = P(D and G) / P(D) = 160/200 = 0.80 (80%).

(d) Prioritizing further testing. The baseline disease rate is P(D) = 0.20. Among patients with elevated glucose the observed disease rate is 57.1%, about 2.9 times the baseline, so these patients merit earlier confirmatory testing. The same numbers also show a limitation: P(G | D) = 0.80 means 20% of diseased patients have normal glucose and would be missed by a glucose-only rule. The threshold should be set using sensitivity, specificity, predictive values, the cost of missed disease versus false alarms, and validated clinical guidance. The association is sample-specific and does not establish causation.

Task 4 - Bayesian Statistics for Disease Classification (4 marks)

(a) Given P(D) = 0.10, P(+ | D) = 0.90 and P(+ | not D) = 0.05, so P(not D) = 0.90.

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P(+) = P(+|D)P(D) + P(+|not D)P(not D) = 0.90(0.10) + 0.05(0.90) = 0.09 + 0.045 = 0.135
P(D | +) = P(+|D)P(D) / P(+) = 0.09 / 0.135 = 2/3 = 0.667 (66.7%)
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Frequency check. In 1,000 people: 100 have the disease and 90 test positive; 900 do not and 45 test positive (5%). So 135 tests are positive, of which 90 are true positives, giving 90/135 = 66.7%.

(b) Roles of the terms. The prior P(D) = 0.10 is the disease probability before the test. The likelihood P(+ | D) = 0.90 is how strongly the evidence is expected under disease. The evidence P(+) = 0.135 normalizes across both groups. The posterior P(D | +) = 0.667 is the prior multiplied by the likelihood, then rescaled. Even a 90% sensitive test yields only a 67% posterior because the disease is uncommon, so false positives from the large healthy group are substantial.

(c) Why Bayesian reasoning helps. Medical information is uncertain and incomplete. Bayes' theorem combines a prior that varies between populations (a screening clinic versus a specialist clinic) with test evidence, and it updates sequentially as new results arrive. It also makes the assumptions explicit. Its usefulness depends on well-calibrated priors and on modelling correlated tests properly, because treating dependent tests as independent overstates the evidence.

Task 5 - Entropy and Information Gain (3 marks)

(a) Entropy. p(high) = 40/100 = 0.4 and p(low) = 60/100 = 0.6.

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H(Y) = -[0.4 log2(0.4) + 0.6 log2(0.6)] = -[0.4(-1.3219) + 0.6(-0.7370)] = 0.5288 +
0.4422 = 0.971 bits
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(b) Meaning. Entropy measures impurity. For two classes it ranges from 0 (all one class) to 1 bit (50:50). A value of 0.971 means the records are very mixed, so a randomly chosen record is almost as uncertain as possible.

(c) Information gain. IG(Y, X) = H(Y) - sum over v of P(X = v) H(Y | X = v). Suppose a glucose threshold splits the 100 records into a high-glucose group (50 records: 32 high-risk, 18 low-risk) and a normal-glucose group (50 records: 8 high-risk, 42 low-risk).

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H(high group) : p = 0.64, 0.36 -> 0.943 bits H(normal group) : p = 0.16, 0.84 -> 0.634
bits Weighted entropy = 0.5(0.943) + 0.5(0.634) = 0.789 bits IG = 0.971 - 0.789 = 0.182
bits
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The split reduces uncertainty by 0.18 bits, so glucose is a useful feature here. Caveats: the threshold must be chosen on training data and checked on held-out data; information gain favours features with many possible values; and it measures association, not causation or clinical value.

Task 6 - Numerical Computation and Distance Measurement (3 marks)

Normalized features in the order glucose, blood pressure, cholesterol: A = (0.70, 0.80, 0.60) and B = (0.50, 0.75, 0.90).

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d(A,B) = sqrt[(0.70-0.50)^2 + (0.80-0.75)^2 + (0.60-0.90)^2] = sqrt[0.0400 + 0.0025 +
0.0900] = sqrt(0.1325) = 0.364
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(a) The Euclidean distance is 0.364. Cholesterol contributes most of the squared difference (0.09 of 0.1325).

(b) Similarity. A smaller distance means the two records are closer in the chosen feature space. Nearest-neighbour methods and K-means rely on this. It does not mean the two patients share a diagnosis: family history, medication and symptoms are not in the vector, and the result depends on the scaling, weights and metric chosen.

(c) Why normalization is needed. Raw glucose varies by tens or hundreds of mg/dL, while blood pressure may vary by only a few mmHg. Without scaling, the largest-unit feature decides the distance. After scaling both features lie in [0, 1], so their differences are comparable. Scaling statistics should come from the training set only.

Task 7 - Bias-Variance Trade-off (4 marks)

(a) Model A (high training and validation error) has **high bias** and is underfitting. Model B (very low training error, high validation error) has **high variance** and is overfitting. Model C (low and similar errors) shows good generalization, assuming the validation data are representative.

(b) Complexity. Increasing complexity lowers bias and raises variance. Expected test error is commonly decomposed as bias^2 + variance + irreducible noise. A too-simple model misses real structure such as nonlinear age-glucose interactions; a too-flexible model memorizes sample-specific noise. The best model lies between these extremes.

(c) Improvements.

- **Model A (high bias):** add informative features or nonlinear terms; use a more flexible model; reduce regularization; check preprocessing for lost signal; train longer if optimization stopped early.
- **Model B (high variance):** collect more representative data; apply L2 regularization, dropout or pruning; simplify the model; use early stopping and cross-validation; check for leakage between training and validation sets.

Task 8 - Overfitting, Underfitting and Model Capacity (4 marks)

(a) Classification. The simple model that fails to capture patterns is **underfitting**: both training and validation errors are high. The moderately complex model that performs consistently is **appropriate generalization**. The highly complex model that is near-perfect on training but poor on unseen patients is **overfitting**.

(b) Capacity. Capacity is the set of functions a model can represent. A linear model cannot represent interactions between age and glucose; a deeper tree or neural network can. Capacity is only useful when it is matched by enough representative data and suitable regularization.

(c) Why performance improves then declines. At low capacity, extra flexibility captures real structure, so training and validation errors both fall. Beyond the right level, the model starts fitting measurement noise, data-entry artifacts or site-specific shortcuts. Training error keeps falling, validation error rises, and the gap widens. The right capacity is where validation error is lowest, which is found with validation curves rather than training accuracy.

Task 9 - Supervised Learning and Model Evaluation (3 marks)

(a) Why supervised. Historical records carry known labels (diseased or disease-free). The algorithm learns a mapping from features X to labels y and then predicts for new patients. Logistic regression, decision trees, random forests, SVMs and neural networks are all suitable, depending on data size, interpretability and calibration needs.

(b) Training and testing. The training set fits model parameters. The validation set selects hyperparameters and thresholds. The test set is used once, at the end, to estimate performance on unseen patients. Splits should be made at the patient level so that repeated records of one patient do not appear on both sides.

(c) Accuracy.

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Accuracy = correct predictions / total test predictions = 92 / 100 = 0.92 (92%)
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The model classified 92 of 100 test patients correctly and made 8 errors. Accuracy alone can mislead: if only 5% of patients have the disease, predicting disease-free for everyone gives 95% accuracy while detecting nobody. Sensitivity, specificity, precision and calibration should also be reported.

Task 10 - Unsupervised Learning and Patient Grouping (4 marks)

(a) Why unsupervised. The records have features but no risk labels. Clustering can reveal groups of patients with similar profiles that may merit further clinical study, without assuming what the groups mean.

(b) K-means procedure. K-means partitions the patients into K clusters by minimizing the within-cluster sum of squares:

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J = sum over i of ||x_i - mu_(c_i)||^2 (c_i = cluster of patient i; mu_k = mean vector of cluster k)
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- Choose K and initialize centroids (for example with k-means++).
- Assign each patient to the nearest centroid.
- Recompute each centroid as the mean of its assigned patients.
- Repeat until assignments stop changing. K-means can reach local minima, so run several seeds and compare.

(c) Distance as similarity. Assignments use Euclidean distance in the scaled feature space. Smaller distance means more similar profiles. Without scaling, the largest-unit feature defines the clusters. K can be guided by the elbow of J and the silhouette score.

(d) Supporting clinical investigation without replacing diagnosis. Summarize each cluster by means, medians and ranges; check that clusters are stable across resamples, sites and time; ask clinicians whether a profile suggests useful questions; avoid treating post-hoc outcome differences as causal; and check that clusters do not reflect unequal access to care or data-collection practices. Clusters generate hypotheses only.

Task 11 - Comparison of Supervised and Unsupervised Learning (4 marks)

(a) Paradigm. Dataset A has features and disease labels, so **supervised classification** is appropriate: the target is known for training examples. Dataset B has no labels, so **unsupervised methods** (K-means, hierarchical clustering, dimensionality reduction) are appropriate. Dataset B cannot directly teach a disease-status target.

(b) Objectives and outputs.

Dimension	Supervised (Dataset A)	Unsupervised (Dataset B)
Input	Features X with labels y	Features X only
Objective	Predict a defined outcome	Discover structure or representation
Output	Class, probability, score	Clusters, embeddings, latent factors
Evaluation	Held-out metrics, calibration	Stability, internal indices, clinical usefulness
Main risk	Overfitting, leakage, label bias	Arbitrary or unstable groups; over-interpretation

(c) Recommendation for a new patient. Use a **supervised classifier** trained on reliable labelled records. It directly optimizes the prediction target and can be checked against known outcomes. Clustering alone cannot say which group means disease. A sensible hybrid is: train and calibrate the classifier; run clustering as separate exploration; compare clusters with outcomes; and confirm reproducibility and clinical value before use.

Task 12 - Design Challenge: Selecting and Adapting a Learning Strategy (8 marks)

Decision: modify the original approach into a two-branch pipeline. Keep the supervised model as a baseline, and add an unsupervised branch and stronger validation.

A. Impact analysis (2 marks).

- **(a) Size and computation.** Storing 1,000,000 records with 4 float64 features takes 1,000,000 x 4 x 8 = 32,000,000 bytes, about 32 MB, before labels, copies and intermediate arrays. Work grows roughly linearly with n. A linear model costs O(nd) per pass, and K-means costs about O(nKdI). A 100-fold increase in records means roughly 100 times the work per pass, so mini-batch training, efficient data loaders and profiling become necessary.
- **(b) Quality and variability.** Normalization stops scale dominance and stabilizes optimization. Missing values must be quantified by feature and site, then imputed with statistics learned from training data only; missingness that depends on clinical concern may carry signal but also workflow bias. Larger, more varied

data exposes unit mismatches, duplicates, population shift and inconsistent labels, so validation checks are needed at ingestion.

B. Evaluation of the original approach (2 marks).

- **(a) One supervised method is not enough.** A classifier optimizes a predefined label. It does not automatically reveal patient groups, especially when group structure differs from the disease boundary. Prediction and discovery are distinct objectives and need separate analyses.
- **(b) Limits of labelled learning.** Labelled records may cover only tested patients, certain sites or care pathways. Labels may be delayed or noisy. Useful structure in unlabelled records is ignored. Semi-supervised learning can help, but pseudo-labels can reinforce errors, so it must be compared with a supervised baseline on independent labelled data.

C. Adaptation or redesign (2 marks).

Stage	Method / decision	Reason
1. Ingest and validate	Schema, range and duplicate checks; unit harmonization	Prevents avoidable errors at scale
2. Split	Patient-level, and temporal or site-level if deployment requires	Avoids leakage; tests realism
3. Preprocess	Imputation and scaling fitted on training data; missingness indicators if justified	Consistent representation
4. Supervised branch	Logistic regression baseline; compare gradient-boosted trees or MLP	Estimates P(D = 1 | x)
5. Calibration and threshold	Calibration curve, Brier score; threshold chosen for clinical utility	Meaningful risk probabilities
6. Unsupervised branch	K-means (or mini-batch K-means) on scaled features, labelled and unlabelled records	Finds groups for hypotheses
7. Evaluation	Held-out metrics; cluster stability and profile review	Checks performance and reproducibility
8. Monitoring	Drift, calibration, missingness and subgroup tracking	Detects deterioration after launch

How the mathematics is incorporated: the supervised branch estimates P(D = 1 | x) and is calibrated; distance (Euclidean on properly scaled features, or a metric suited to mixed data) drives clustering; normalization parameters come from training data only and are reused in production; and model capacity starts small (for example logistic regression), increasing only when validation shows benefit.

D. Experiments and evidence-based defence (2 marks).

(a) Experiment plan. Use one fixed, leakage-safe split. Fit a logistic regression baseline with documented imputation and scaling; compare regularized alternatives (tree ensemble, small MLP) only if they improve validation results. Run ablations on scaling, imputation and missingness indicators. Benchmark runtime and memory at 10k, 100k and 1M records. For clustering, compare candidate K values, several seeds, silhouette score, within-cluster sum of squares and bootstrap stability. Test robustness on held-out sites or time periods.

Evaluation area	Measures
Predictive	ROC-AUC, PR-AUC, sensitivity, specificity, PPV, Brier score, calibration plot, confusion matrix
Clustering	WCSS, silhouette score, adjusted Rand index across resamples, cluster-size balance, clinician review
Operational	Training time, peak memory, throughput at 1,000,000 records

(b) Justification. The hybrid design assigns each objective to a suitable paradigm rather than forcing one algorithm to serve incompatible goals. Mathematically, scaling makes distances meaningful, calibration makes probabilities interpretable, and the capacity search balances underfitting against overfitting.

Computationally, linear and mini-batch methods scale to one million rows. Generalization is protected by patient-aware splits, training-only preprocessing, held-out and external validation, and monitoring. Use clusters for investigation, not diagnosis, and deploy the risk model only after clinical validation and governance review.

Overall Conclusion

Feature scaling, probability (conditional and Bayesian), entropy, distance, bias-variance and supervised versus unsupervised reasoning all determine how patient data are represented and whether a model can be trusted. The worked results include P(D | G) = 57.1%, P(D | +) = 66.7%, entropy 0.971 bits, information gain 0.182 bits, distance 0.364 and a classification accuracy of 92%. For one million records, a hybrid pipeline with calibrated supervised prediction, clustering for exploration, and independent validation is the recommended design.

Final principle. A model should support, not replace, clinical judgement. Reliable deployment needs representative data, validated probabilities, clinically appropriate thresholds, fairness and safety review, privacy protection and ongoing monitoring.